a. Prior-Induced Confounding Bias

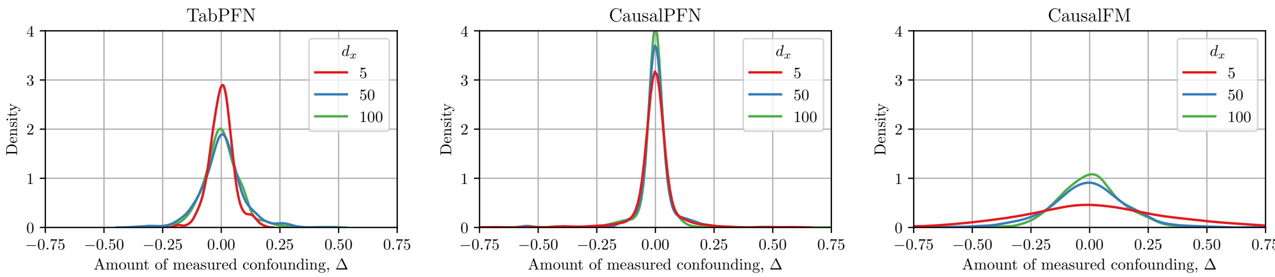

Figure 1. Prior-induced confounding bias of different PFNs (updated). High values of Δ correspond to a high degree of the observed confounding. Thus, when Δ is concentrated around zero, the prior might induce the confounding bias which does not vanish asymptotically with growing data (as a strong observed confounding is excluded a priori). Here, we sample $B = 512$ causal datasets with $n = 10000$ each.

b. Detailed ACIC-2016 results

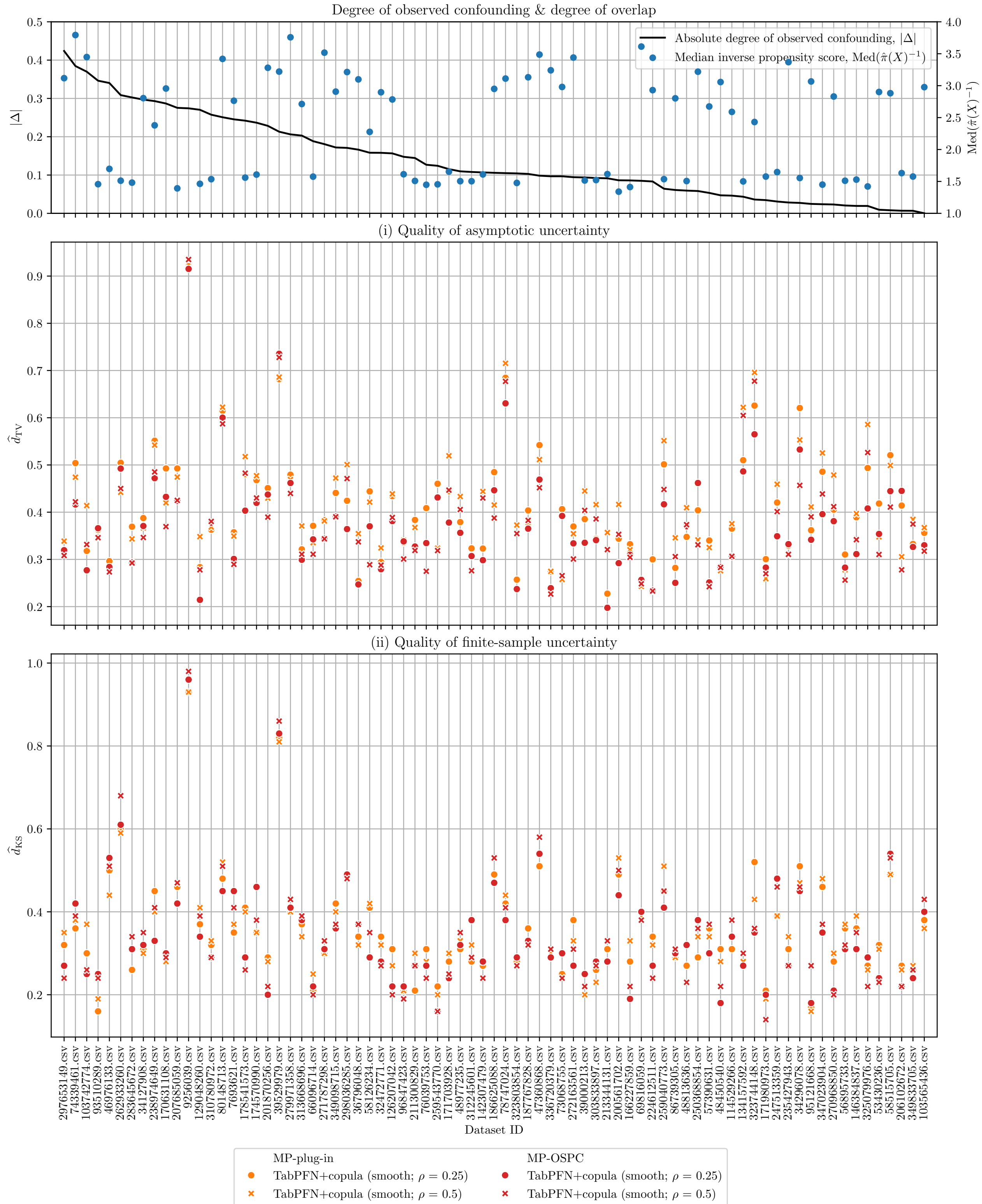

Figure 2. Full (i) asymptotic and (ii) finite-sample results of ATE estimation with TabPFN-based estimators for 77 semi-synthetic ACIC 2016 datasets. Datasets are sorted wrt. the absolute degree of measured confounding, $|\Delta|$; and the median inverse propensity score is provided for reference. Reported: (i) mean $\hat{d}_{\mathrm{TV}}$ and (ii) $\hat{d}_{\mathrm{KS}}$ over 10 runs (lower is better). **Results.** Our MP-OSPC is especially effective for $|\Delta| > 0.25$, namely when the actual degree of measured confounding is outside of the TabPFN's prior support.